



CS355 – Database Project Proposal Form

Section 1: Metadata

to be filled by the student

1.1. Project Information to be filled by the student

Title - FitCore: Fitness Tracking and Trainer Management Ecosystem	
Section:	Instructor:

1.2. Student(s) Information

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Submission guideline: Save your project proposal as a pdf file and rename as Project Proposal_L1_ProposedTitle where L1 is to be replaced with your section



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Section 2: The Project

to be filled by the student

2.1. Project Description: *Please provide a brief introduction of the project including its scope.*

Introduction

The FitCore project is a robust fitness management system designed to enhance user engagement with fitness programs while providing specialized tools for clients, trainers, and administrators. The platform offers a secure registration and authentication process, allowing users to sign up as clients, trainers, or admins, each with unique login credentials and verification procedures.

Purpose and Functionality

FitCore's primary purpose is to streamline the management of fitness activities and user interactions. Clients can log workouts, set and track fitness goals, and receive personalized guidance from trainers. Trainers can create and manage workout plans, monitor client progress, and provide feedback. Administrators have the ability to manage user and trainer accounts, monitor system activity logs, and handle financial transactions, ensuring the platform operates smoothly and securely.

Scope

FitCore supports three main user roles:

- **Clients:** Users who log their workouts, set goals, track progress, and receive guidance from trainers.
- **Trainers:** Fitness professionals who create detailed workout plans, monitor client progress, and interact with clients for personalized coaching.
- **Admins:** System administrators who manage user accounts, oversee system activity, and handle financial aspects to ensure platform efficiency.



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2.2 Functional Requirements

This section describes each function/feature provided by your system. These functions are logically grouped into modules based on their purposes. The users in your system must be categorized such as client, customer or administrator etc. These users will be accessing the database with the level of access that they are authorized with.

Module 1: User Registration and Authentication

Function 1: Register an account

The system lets users register an account on the website as a client, trainer, and admin.

– Function 1a: Register as a client The register form prompts the user to enter their details i.e. Name, Email, Password. The form is submitted and an unverified client account is created. The user receives a link on their email address which completes account verification. The client is then given a User ID.

– Function 1b: Register as a trainer The register form prompts the user to enter their details i.e. Name, Email, Password, Experience, Qualification. The form is submitted and an unverified customer account is created. The user receives a link on their email address which completes account verification. The trainer is then given an Employee ID.

– Function 1c: Register as an admin The register form prompts the user to enter their details i.e. Name, Email, Password. The form is submitted and an unverified customer account is created. The user receives a link on their email address which completes account verification. The admin is then given an Employee ID.

Function 2: User Login

The system lets users login as a client, trainer or admin.

- Function 2a: The client must input their User ID and password to login to the system each time.
- Function 2b: The trainer must input their Employee ID and password to login to the system each time.
- Function 2c: The admin must input their Employee ID and password to login to the system each time.

Module 2: Fitness Tracking by User

Function 1: Log Workout

Users can log their daily workout by entering details like workout type, duration, calories burned, and additional notes. The system stores this information in the user's workout history.

Function 2: Track Progress

Users can view their workout history, analyze trends in their fitness progress (e.g., weight loss or muscle gain), and get statistics on total calories burned, time spent exercising, etc.

Function 3: Set Goals

Users can set fitness goals (e.g., weight loss, distance running, etc.). The system tracks their progress toward these goals by comparing daily workout logs against the set objectives.



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Function 4: Trainer Interaction

Users can request guidance from Trainers. Trainers can view Users' progress, set custom workout plans, and provide feedback through direct messages or comments on workout logs.

Module 3: Trainer Features

Function 1: Create Workspace Plans

Trainers can create detailed workout plans that include exercise names, sets, reps, and instructional videos. These plans can be assigned to individual users.

Function 2: View User Progress

Trainers can access the progress history of assigned users, offering personalized advice based on tracked workout data.

Function 3: Manage Clients

Trainers can manage their client list, track client activities, and communicate with them for better guidance.

Module 4: Admin Features

Function 1: Manage Users and Trainers

Admins can approve or reject Trainer registrations and manage User and Trainer accounts (e.g., deactivating accounts)

Function 2: Monitor System Activity

Admins have access to logs showing system activity, user signups, and potential issues.

Function 3: Manage Finances

Admins are responsible for overseeing the financial aspects of the platform. Admins can view transaction histories, generate financial reports, and ensure that all monetary transactions are conducted securely. They can also manage pricing for services, such as workout plans, and monitor the financial health of the platform by tracking income and expenses.

2.3. Planned Schedule: *Kindly list the start/end dates and the timeline for the achievement of any intermediate milestones and the expected contribution to be made by the participant(s).*



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Week 7	Finalize the system's Entity-Relationship Diagram (ERD)
Week 8	Research common fitness metrics and compile the necessary datasets
Week 9	Develop the back-end functions and database
Week 10	Work on the graphical user interface (GUI) and front-end development
Week 11-12	Implement SQL for data management
Week 13	Implement transaction management to ensure data integrity
Week 14	Perform final checks and add any additional features

2.4. Technology Stack: *If you are utilizing any language or database besides PyQt and SQL Server, please complete this section; otherwise, leave it blank. Specify the programming language and database management system intended for constructing this application, as well as the application type (Desktop, Web, or Mobile).*

Softwares:

1. DBdesigner
2. MySQL
3. Visual Studio Code
4. GitHub

Programming Languages:

1. Python
2. SQL



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2.5. Screens: Provide images of all application screens, showcasing clear input and corresponding outputs. Ensure each image includes a concise caption explaining user action and expected/observed output. You can create these screens using Qt Designer.

Qt MainWindow - save.ui*

Type Here

LOGIN

Sign in to your existing account

User ID:

Password:

User Type

☐ Client ☐ Admin ☐ Trainer

LOGIN SCREEN



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A screenshot of a Qt application window titled "MainWindow - Registration.ui*". The window contains a registration form with the following elements:

- A title bar with a Qt logo and window control buttons (minimize, maximize, close).
- A text input field at the top with the placeholder text "Type Here".
- A large heading "REGISTER" in bold, black, uppercase letters.
- A subtitle "Create a new account" in a smaller, black, sans-serif font.
- Form fields for "First Name:" and "Last Name:" with white text boxes.
- Form fields for "Password:" and "Confirm Password:" with white text boxes.
- An "Email:" label followed by a white text box.
- A "User Type" section with two radio buttons: "Client" and "Trainer".
- Two buttons at the bottom: "Register" and "LogIn", both with black text on a light gray background.

REGISTRATION SCREEN



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Qt MainWindow - save2 - Copy.ui*

Type Here

Salman Kamal
Male, 30 years

HEIGHT 115 cm **WEIGHT** 78 kg

Upcoming Meetings

Morning Meeting:

- Discuss project timeline
- Review system architecture

Afternoon Meeting:

- Team progress check-in
- Address any roadblocks

Day 2, Week 6
10/09/2024

	Client Name	Workout Plan	Progress	Assigned Date
1	Minhaj ul Hassan	Fat Loss	0%	Aug 1, 2024
2	Zarrar ul Arfeen	Muscle Gain Program	100%	Sept 4, 2024
3	Hassan Shahzad	Full Body Toning	40%	Sept 4, 2024
4	Aitzaz Rahim	Cardio and Body	40%	Sept 4, 2024
5	Kazi Don	Strength Training	80%	Sept 10, 2024
6	Haris Khan	Muscle Gain	10%	Sept 21, 2024

[Schedule a meeting](#) [Contact Client](#)

[Log Out](#) [Switch Account](#)

TRAINER SCREEN



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Qt MainWindow - untitled 9*

Type Here

Zarrar ul Arfeen
Male, 19 years

HEIGHT 185 cm **WEIGHT** 12 kg

MEALS

Dinner:
- Milk
- Brown Bread
- Eggs

Afternoon Snack:
- Vegetable Salad

Lunch:
- Biryani
- Brown Bread
- Eggs

Day 2, Week 6
10/09/2024

Trainer
Salman Kamal

	Activity	Sets	Reps	Cal. Burn
1	Running	-	-	300
2	Push-ups	3	15	100
3	Squats	4	12	180
4	Cycling	-	-	450
5	Deadlifts	4	10	200
6	Jump Rope	3	12	300
7	Plank	-	-	80

Log Out Switch Account

Workout Log Contact Trainer

CLIENT SCREEN



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Qt MainWindow - untitled 6*

Type Here

WORKOUT LOG

	Activity	Time	Dist.	Sets	Reps	Cal. Burn
1	Running	30 min	5.0km	-	-	300
2	Push-ups	15 min	-	3	15	100
3	Squats	20 min	-	4	12	180
4	Cycling	45 min	12.5 km	-	-	450
5	Deadlifts	25 min	-	4	10	200
6	Jump Rope	10 min	-	3	12	300
7	Plank	10 min		-	-	80

WORKOUT LOG OF CLIENT



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Admin

Muneeb Siddique

REQUEST

13

DUES

5

Notifications:

- Minhaj ul Hassan (Client) has an overdue payment of Rs. 3000.
- Kazi Don (Trainer) has salary due for August 2024.
- Client check-in missed for Salman Kamal on Sept 5, 2024.
- Kazi Don (Trainer) has salary due for August 2024.

Log Out

Switch Account

Check-in/out times:

	ID	Date	Check-in	Check-out	Type
1	CL1023	Sept 5, 2024	08:00 AM	08:15 AM	Client
2	CL4301	Sept 5, 2024	08:15 AM	08:30 AM	Client
3	TR1234	Sept 5, 2024	09:00 AM	09:15 AM	Trainer
4	TR3452	Sept 6, 2024	08:45 AM	09:15 AM	Trainer

Payment Dues:

	Client ID	Amount Due	Due Date	Status
1	CL1023	Rs. 3000	Sept 5, 2024	Paid
2	CL4301	Rs. 3500	Sept 5, 2024	Paid
3	CL1234	Rs. 3000	Sept 5, 2024	Unpaid

Trainer Salary:

	Trainer ID	Hourly Rate	Total Hours	Amount Due	Status
1	TR1023	Rs. 300	150	Rs. 45000	Paid
2	TR4301	Rs. 350	160	Rs. 56000	Paid
3	TR1234	Rs. 300	140	Rs. 42000	Unpaid

ADMIN SCREEN